

AIMS
Airbus Industrie Material Specification

Aluminium alloy (2024)
Solution treated and naturally aged (T42)
Clad sheet and strip
 $0,2 \text{ mm} \leq a \leq 6,0 \text{ mm}$
Standard surface quality
Material Specification

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1 Scope

This Material Specification defines the requirements of aluminium alloy 2024 T42 clad sheet and strip $0,2\text{ mm} \leq a \leq 6,0\text{ mm}$ standard surface quality, for aerospace application. The products made out of F- and O-temper with a thickness $a > 0,8$ can be in chemical milling quality or “no” chemical milling quality, see ABS 5044. This specification does not apply to delivery condition T3 and T351.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

Not applicable.

3 Normative references

This Airbus Industrie specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus Industrie specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred shall be applied.

AIMS 03-04-000	Airbus Industrie Material Specification Technical Specification Aluminium and aluminium alloy wrought products Sheet, plate and strip up to 12 mm
AIMS 03-04-009	Airbus Industrie Material Specification Material Specification Aluminium alloy (2024) Solution treated, controlled stretched and naturally aged (T351) Clad sheet $0,8 < a \leq 6,0\text{ mm}$ – standard surface quality – Chemical milling quality
AIMS 03-04-014	Airbus Industrie Material Specification Material Specification Aluminium alloy (2024) Solution treated and naturally aged (T3) Clad sheet and strip $0,2\text{ mm} \leq a \leq 6,0\text{ mm}$ standard surface quality
EN 2032-2	Aerospace series – Metallic materials – Part 2 Coding of the metallurgical condition in delivery condition
EN 4500-2	Aerospace series – Metallic materials Rules for drafting and presentation of material standards Part 2: Specific requirements for aluminium, aluminium alloys and magnesium alloys
ABS 5044	Aerospace series / Clad sheets and plates / standard surface quality Aluminium alloy 2024 Thickness $0,2 \leq a \leq 12\text{ mm}$, Dimensions

4 Technical requirements

4.1 General

Specification values are individual minimum except where stated.

4.2 Specific

Data requirements shall be given in tables according to the relevant material part of EN 4500-2.

5 Designation

Refer to ABS 5044.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 03-04-000.

1	Material designation		Aluminium alloy 2024										
2	Chemical composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	others		Al
											each	total	
		min.	-	-	3,8	0,30	1,2	-	-	-	-	-	-
		max.	0,50	0,50	4,9	0,9	1,8	0,10	0,25	0,15	0,05	0,15	Base
3	Method of melting		-										
4	Form Method of production Limit dimension (mm)		Clad sheet and strip Rolled 0,2 ≤ a ≤ 6,0										
5	Technical specification		AIMS 03-04-000										

6	6.1 Delivery condition and heat treatment	F ¹⁾ As fabricated	O ¹⁾ Annealed	T351 ²⁾ purchased against AIMS 03-04-009	T3 ²⁾ purchased against AIMS 03-04-014
	6.2 Delivery condition code ³⁾	F	A	K	
7	Use condition and heat treatment	T42 Delivery condition 490°C ≤ Θ ≤ 500°C/WQ Θ ≤ 40°C and naturally aged Θ = ambient/t ≥ 5d			

8	Test sample Test piece Heat treatment			Delivery condition: 0		Use condition : T42 ⁴⁾			
9	Dimensions concerned	a	mm	0,2≤a≤1,6	1,6<a≤6,0	0,2≤a≤0,8	0,8<a≤1,6	1,6<a≤6,0	
10	Thickness of cladding on each face		%	4	2	4		2	
11	Direction of test piece			L or LT	L or LT	L or LT		L or LT	
12	Tensile	Temperature	Θ	°C	Room temperature				
13		Proof stress	R _{p0,2}	MPa	≤ 100	≤ 100	235	235	250
14		Strength	R _m	MPa	≤ 210	≤ 220	395	395	415
15		Elongation	A ₅₀	%	12	12	12	15	15
16		Reduction of area	Z	%					
17	Hardness		HB	-					
18	Shear strength		R _c	MPa					
19	Bending		k	-	0,5	1	-		
20	Impact strength			--					
21	CREEP	Temperature	Θ	°C	--				
22		Time		h	--				
23		Stress	σ _a	MPa	--				
24		Elongation		a	%	--			
25		Rupture stress		σ _R	MPa	--			
26		Elongation at rupture		A	%	--			
27	Notes (see line 98)			1) 2) 3) 4) 5) 6)					

44	External defects		-	See AIMS 03-04-000 Table 8							
62	Diffusion in the cladding ^{4) 6)}		-	See AIMS 03-04-000							
			7	Thickness (mm)	0,4 ≤ a <0,6	0,6 ≤ a < 0,8	0,8 ≤ a < 1,0	1,0 ≤ a < 1,2	1,2 ≤ a ≤ 1,6		
			Allowable diffusion depth of actual cladding thickness in %	80 [tbd]	55 [tbd]	45 [65]	35 [50]	30 [40]			
			Thickness (mm)	1,6 < a <2,0	2,0 ≤ a < 2,5	2,5 ≤ a < 3,2	3,2 ≤ a ≤ 6,0				
			Allowable diffusion depth of actual cladding thickness in %	35 [75]	30 [65]	25 [50]	20 [40]				
64	Surface condition roughness ^{4) 5)}		-	See AIMS 03-04-000 chapter 4.2.2.7							
			7	Thickness (mm)	0,8 < a ≤ 4,0			4,0 < a ≤ 6,0			
			Mean roughness value Ra (µm)	4,0			6,3				
72	Cladding chemical composition %	Element	Si + Fe	Cu	Mn	Mg	Zn	Ti	others		Al
		min.	-	-	-	-	-	-	each	total	
		max.	0,7	0,10	0,05	0,05	0,10	0,03	0,03	-	Base
98	Notes		-	1) For delivery in F condition the test piece heat treatment is T42, for delivery in O condition the test piece heat treatment is O and T42 2) For delivery in T3 condition the requirements of AIMS 03-04-014 and for delivery in T351 condition the requirements of AIMS 03-04-009 are valid and this specification does not apply 3) See EN 2032-2 4) Testing in condition T42 with stretching 0,5 to 1 %, if necessary 5) Not applicable for sheet in O- or F-condition for “no” chemical milling (see ABS) 6) The values in brackets [] belong to O-condition heat treated to T42							